

SCALE AND RESILIENCE REPORT

EVERY NUMBER, WITH THE MACHINE IT CAME FROM

This is the canonical record behind every performance figure on this site. Each result names its rig, its method and the command that reproduces it. Where two captures disagree, the later quiet-box gate is authoritative and the earlier one is kept for history. Publish methodology, not just numbers.

ENVIRONMENT

THE THREE RIGS

Nothing here ran on cloud hardware. These are the actual machines. Bigger hardware moves the ceilings; the report states floors.

RIG	CPU	RAM	OS AND KERNEL	JVM	BROKER	USED FOR
Linux reference rig	Intel Core i7-9700T, 6 cores, 6 threads, 2.0 GHz	8 GB	Ubuntu 26.04 LTS on WSL2, kernel 6.18	OpenJDK 17.0.20	Embedded HiveMQ CE 2026.5, 4 shared consumers	2026-08-15 quiet-box gate: capacity floor, offered ramp, latency.

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RIG	CPU	RAM	OS AND KERNEL	JVM	BROKER	USED FOR
Mac bench gateway	x86_64, 4 logical / 2 physical cores	not recorded	macOS 12	Temurin JDK 17	HiveMQ CE 2026.5 on loopback	Gap-closure capture (projected end-to-end, latency, store-and- forward), TCK session, standby takeover, live gateway projection on Ignition 8.1.38
Mac 8- core dev box	8 cores, model not recorded	not recorded	macOS	JDK 17	Embedded Moquette, all in one JVM	R3 gate 2026-08-11: codec, single-edge saturation, soak, fan-in, multi-broker

All-in-one-JVM runs put edge, broker and host in one process, so the numbers exclude network round trips and include GC contention between roles. The live gateway rows run the shipped modules on a real Ignition 8.1.38 gateway against a separate HiveMQ CE process.

METHODOLOGY

HOW THE LOAD IS SHAPED



PARAMETER	SETTING
Sites (edge nodes)	One edge per 50,000 wire metrics: 1, 5 and 10 edges for the 50k, 250k and 500k tiers; 20 edges on the opt-in 1M rig. One device BIRTH per edge
Metrics	50,000 / 250,000 / 500,000 wire metrics resident in Host state; 1,000,000 opt-in
Update rates	Capacity: a ramp that raises offered load step by step until delivered rate stops tracking offered, so the knee is found rather than asserted. Latency: paced at about 10,000 messages/s so the path is measured unsaturated. Fan-in: 600,000 updates/s offered throughout
Payload	1,000 metrics per NDATA in the capacity ramp; 50 metrics per message in the codec test; 1 metric per message in the latency test. Sparkplug B protobuf, alias-only after BIRTH
Projection cost	Real per-leaf tag-write cost injected at 50 microseconds per leaf so the end-to-end figure includes writing values into tags, not just decoding them
Duration	Warm-up phase discarded before every measurement (a cold JIT reads as latency on any code). Steady windows of 5 s per ramp step; 20 s soak for heap; node-churn soak configurable to 30 minutes or more
Quiet box	Nothing else running. A concurrent Gradle test JVM halves the 500k tier, which is why every published gate says quiet-box

RESULTS

RESULTS BY DIMENSION

The headline row per dimension is the one quoted elsewhere on the site. Earlier captures are listed so the trend is visible.



DIMENSION	RESULT
Sustained capacity, projected through to visible tags	at least 339,527 updates/s, lowest of the 50k, 250k and 500k tiers 44,769 updates/s at 250k, 4 projector shards; 19,507 at 50k, 1 shar 341k / 340k / 340k across tiers
Offered ramp absorbed without bending	480,000 updates/s, ramp exhausted, ceiling not found
End-to-end latency, edge to broker to host, warm and paced	p50 0.427 ms · p99 0.933 ms p50 0.216 · p95 0.366 · p99 0.517 · p99.9 1.489 · max 3.576 ms, 40,
Host resident state	500,000 wire metrics in under 4 GiB; about 300 bytes per metric per

DIMENSION

RESULT

Wire ingest,
no projection

59,980 / 59,910 / 59,902 updates/s at 50k / 250k / 500k, 0 loss

Codec
throughput,
single thread

3.9M metric-updates/s, encode plus decode round trip

Store-and-
forward
replay after a
broker
outage

40,035 updates/s, loopback

400,000 of 400,000 replayed in 20.33 s = 19,671 updates/s, zero los

Outage
buffering,
durable

2,697 messages/s with fsync every 32 records; about 90 messages/s w

Host scale-
out with
MQTT 5
shared
subscriptions,
ingest only

11,315 (1 broker x 1 consumer) to 222,000 (1 x 4) updates/s, about



DIMENSION

RESULT

Multi-broker scale-out, 16-edge fleet 211 to 215,695 updates/s delivered across 8 brokers against 1, about

Single edge, saturated about 100,000 updates/s, 0 loss

Soak, 20 s about 215,000 updates/s, heap flat, leak-free

Warm-standby Host takeover **244 ms** from offline STATE to promoted online STATE, 25 nodes, 200 m

Live gateway projection, scalar 20,000 leaves across 4 device BIRTHs ready in 1.03 s, about 20,000

Live gateway projection, templates 1,000 x 15 template members ready in 1.80 s, about 8,300 members/s,

The capacity row is a floor, not a peak. The 2026-08-15 ramp was exhausted at 480,000 updates/s without the run bending, so the ceiling on that rig is unknown and higher. Every capacity claim on this site is "at least", never "up to". The gap-closure figures are lower because that rig has 4 logical cores and the projector shards by core count.



WHAT WAS BROKEN ON PURPOSE

Each row is a test in the shipped suite. The first group runs in ordinary CI on every build; the second is the opt-in load suite.

INJECTED FAILURE	REQUIRED BEHAVIOUR	TEST
Broker killed mid-session, restored later	Edge buffers to disk, reconnects, rebirths, replays in order with source timestamps and the historical flag; zero loss	StoreForwardTest
Disk append and fsync failures while buffering	Counted, surfaced as degraded, recovered when the disk returns; a flush after an open failure does not clear the degraded state	DiskFullStoreForwardTest, DiskBufferPartialWriteRecoveryTest
Garbage payloads, malformed topics, oversized messages, skewed timestamps	Host survives and keeps bridging a legitimate edge	ChaosTest, MalformedStructuredPayloadTest
50,000 mutated protobuf payloads	No crash, no hang	DecoderFuzzTest
Untrusted TLS certificate	Connection rejected	TlsNegativeTest
Publish failures on the live path	Counted and retried without blocking the tag subscription	LivePublishFailureTest, PublishPipelineFlushFailureTest

INJECTED FAILURE	REQUIRED BEHAVIOUR	TEST
Primary Host crash	Standby promotes after debounce; edges do not rebirth	standbyDrill harness, documented in HA.md
Node churn for 30 minutes or more	Heap, thread count and node inventory stay bounded; dead nodes evicted	LongSoakTest (load suite)
5,000 distinct edge nodes on one Host	Bounded heap	LargeFleetTest (load suite)
Rebirth storm from sequence gaps under fan-in	Rate-limited per node; loss at 4 edges fell from 76% to 5% after the fix this test found	LoadTest fan-in scenario

CONFORMANCE

CONFORMANCE IS THE BASELINE

Eclipse Sparkplug TCK 3.0.0, ratified binary, 2026-08-15: 182 Edge and 52 Host assertions passed, zero FAIL lines across ten runs. Host MultipleBroker and the BROKER profile were not run, stated on the [conformance record](#) with the unedited logs.

REPRODUCE



RUN IT YOURSELF

The suite ships in the product repository. Run it on a quiet box and compare against the tables above.

LOAD SUITE

● 4 TO 6 MINUTES

```
./gradlew :common:loadTest --console=plain
./gradlew :common:loadTest --tests io.sparkbridge.perf.CapacityLoadTest
./gradlew :common:loadTest -PloadHeap=9g -Pcapacity1m --tests
io.sparkbridge.perf.CapacityLoadTest
./gradlew :common:gatewayBench -PgatewayBench      # against a live gateway
./gradlew :common:standbyDrill                     # two-JVM takeover
# knobs: -Pshards=N -Dsparkbridge.host.rebirthCooldownMs -
Dsparkbridge.mqtt.maxReconnectMs
```

Results and per-scenario notes are written to the console and to build/reports. Send us yours and we will add the rig to this page.

SIZE YOUR ESTATE FROM THESE NUMBERS

Send site count, tags per site and update rates and we return the sizing with the ceiling it is based on.

[REQUEST AN ARCHITECTURE REVIEW](#)

[BACK TO THE SUMMARY TABLE](#)

[40-SITE REFERENCE ARCHITECTURE \(PDF\)](#)

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

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